

descSPIM-Galaxy Manual: Construction, Operation, and Sample Preparation

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1. Microscope Construction

Stage setup and positioning

Timing 30 min - 1 h

1. Obtain two crossed-roller side-actuated translation stages (XR25P/M, Thorlabs) and loosen each flexure lock to carefully remove the micrometer head.
2. Mount one motorized actuator (Z825B, Thorlabs) onto each translation stage, labeling one assembly as the camera stage and the other as the sample stage.
3. Secure the camera stage (XR25P/M + Z825B) directly to the 300 mm × 450 mm optical breadboard at its designated location.
4. For the sample stage, first secure the sample-holder rail (XR25DR2, Thorlabs) to the breadboard. Then, place the XR25P/M + Z825B assembly on top of it, ensuring that the Z825B cable faces outward, and fix it in place by tightening the M6 holes. Finally, mount the translation stage (XR25/M, Thorlabs) on top with its manual adjustment knob facing outward and secure it by tightening the M6 holes.
5. Position the Cobolt Skyra laser combiner (488, 515, 561, 647 nm; each 50 mW; Cobolt / HÜBNER Photonics) at its designated location on the breadboard ([Fig. 1a](#)).
6. Mount the sample stage onto the rail and tighten the M6 tapped holes just enough to hold it temporarily in place.
7. Route each stage's motor cable to its Kinesis brushed motor controller (KDC101, Thorlabs), plug each controller into its K-Cube and T-Cube power supply (KSP201, Thorlabs), and arrange the controller–power pairs neatly on the breadboard.
8. Bring one single-mode fiber from the on-board Cobolt Skyra combiner to the setup, and position the off-board Cobolt C-FLEX unit (405 nm and 730 nm; 100 mW and 55 mW, respectively; Cobolt / HÜBNER Photonics) so that its fiber pigtail reaches the breadboard without mechanical strain.

CRITICAL STEP: In the following steps, arrange the components in the order of the detection optics, sample holder, and excitation optics. This configuration is essential

because the detection optics in descSPIM-Galaxy are larger than those in the descSPIM-basic system, and the 1X objective lens used in this setup (0.03 NA, TL1X-SAP, Thorlabs) has a shorter working distance (8.0 mm). Therefore, position the camera stage first to avoid physical interference between the sample holder and the excitation optics during assembly.

Constructing of the detection optics

Timing 2 h - 3 h

9. Connect the high-speed motorized filter wheel (FW103/M, Thorlabs) to the stepper motor controller (KST201, Thorlabs) and plug the controller into its dedicated power supply unit (KPS201, Thorlabs), ensuring that all power and communication cables sit without strain before switching on.
10. Remove the filter-wheel cover by unscrewing the retaining screws and insert the filters in order from shortest to longest wavelength: Slot 1, 425 nm long-pass (#84-742, Edmund Optics) for 405 nm excitation; Slot 2, 500 nm long-pass (FELH0500, Thorlabs) for 488 nm excitation; Slot 3, 520/40 nm band-pass (FBH520-40, Thorlabs) for 488 nm excitation; Slot 4, 575 nm long-pass (FELH0575, Thorlabs) for 561 nm excitation; Slot 5, 675 nm long-pass (#84-747, Edmund Optics) for 647 nm excitation; Slot 6, 750 nm long-pass (FELH0750, Thorlabs) for 730 nm excitation.
11. Secure each filter with the provided mounting hardware and replace the cover.
12. Record in your notebook which filter occupies each numbered slot to ensure correct matching of excitation wavelength and detection filter.

CRITICAL STEP: It is essential to record which filter is inserted into each numbered slot. Failure to match the excitation wavelength with the corresponding filter during imaging can result not only in inaccurate data acquisition but also in potential damage to the camera sensor due to direct laser exposure through the detection optics.

13. Connect the 1× objective lens (0.03 NA, TL1X-SAP, Thorlabs) to the cable-feed-outside of the filter wheel using a C-mount-to-SM1 thread adapter (SM1A9, Thorlabs), then replace and secure the filter-wheel cover.

14. On the opposite side of the filter wheel, install the tube lens (TTL200-A, Thorlabs; $f = 200$ mm, ARC 350–700 nm) by threading it into a 10 mm lens tube (SM1L10, Thorlabs) and an SM1-to-SM2 adapter (SM1A2, Thorlabs).
15. Add a second SM1A2 adapter (Thorlabs) and another 10 mm SM1L10 tube (Thorlabs) on the port opposite the tube lens, then mount a kinematic cage cube (DFM1/M, Thorlabs) fitted with a protected silver mirror (PFR10-P01, Thorlabs; 25 mm $\times$ 36 mm) so that its secondary port faces left (right angle to the objective axis).
16. Align the hyperspectral camera (SNAPSCAN VNIR, imec) along the objective axis and the CMOS camera (CS895MU, Thorlabs) perpendicular to it via the cube.
17. Between the cube and the hyperspectral camera, connect the components in the following order from the cage cube toward the hyperspectral camera: a 10 mm SM1 lens tube (SM1L10, Thorlabs), a 5 mm SM1 lens tube (SM1L05, Thorlabs), an SM1 externally threaded adapter for $\varnothing 25$ –25.4 mm ($\varnothing 1$ inch) optics (SM1P2, Thorlabs), and a C-mount-to-SM1 adapter (SM1A9, Thorlabs).
18. On the CMOS side, connect sequentially: 10 mm lens tube (SM1L10, Thorlabs), 5 mm lens tube (SM1M05, Thorlabs), adjustable focus adapter assembly (SM1V10, Thorlabs), and C-mount-to-SM1 adapter (SM1A9, Thorlabs) (**Fig. 1b**).
19. Prepare three posts (TR20/M-JP, Thorlabs; 20 mm, $\varnothing 12$ mm) and three holders (PH30/M, Thorlabs; 30 mm, $\varnothing 12$ mm $\rightarrow$ $\varnothing 12.7$ mm). Use two SM1 series tube clamps (SM1TC, Thorlabs) to secure the front and rear faces of the cage cube.
20. Fix two optical rails (RLA150, Thorlabs; 150 mm) and one optical rail (RLA075, Thorlabs; 75 mm) onto the camera stage, and mount the assembled detection optics on them (**Fig. 1c,d: c, intermediate assembly; d, components arranged on a breadboard**).
21. Finally, connect the external I/O port of the CMOS camera (CS895MU, Thorlabs; 12-pin Hirose HR10A-10P-12S) to the I/O 1 jack of the KDC101 controller (Thorlabs) using a 12-pin Hirose-to-3.5 mm mono TTL trigger cable to enable hardware-level synchronization between stage motion and camera exposure.

CRITICAL STEP: It is essential to tightly secure the 150 mm and 75 mm optical rails to the camera stage in order to ensure mechanical stability.

Construction of the sample holder

Timing 30 min - 1 h

22. Fix the post base (PB1, Thorlabs) to the breadboard with an M6 capscrew and mount the post (P50/M, Thorlabs) to provide a rigid support pillar.

Attach the height-adjustable platform (C1519/M, Thorlabs) to the motorized sample stage using an M6 capscrew.

Mount the manual vertical stage (MVS05/M, Thorlabs) on top of the platform and secure it with an M6 capscrew.

23. Slide the precision rotation stage (PR01/M, Thorlabs) onto the rail dovetail without tightening the clamp.

24. Attach a Ø6 mm cage rod (ER2, Thorlabs; 50.8 mm) to the PR01/M and mount a cage adapter (SP15/M, Thorlabs) onto the rod.

25. Extend four cage rods (ER4, Thorlabs; 101.6 mm) from the SP15/M and secure them with a cage plate (CP31/M, Thorlabs) to form a 30 mm cage column.

26. On the opposite side of the SP15/M, assemble a 16 mm mini-cage using four cage rods (ER1.5, Thorlabs; 38.1 mm) and cap the assembly with either a blank cage plate (SP01/M, Thorlabs) or a cage adapter (SP05/M, Thorlabs).

27. After completing the assembly, lock the fully built PR01/M–SP15/M–cage unit in the designated position on the optical rail (RLA300/M, Thorlabs; 300 mm) using four M6 capscrews.

28. Verify the alignment of the bottom CP31/M plate and the top mini-cage relative to the optical axis and stage travel **(Fig. 1e)**.

29. Place the completed holder assembly on the breadboard and confirm stability with the PB1 + P50/M support pillar **(Fig. 1f)**.

PAUSE POINT:

Construction of the excitation optics

Timing 2 h - 5 h

30. Mount an achromatic collimating lens (AC254-100-A, Thorlabs) into an XY-adjustable holder (CXY1A, Thorlabs) and secure the holder to the center of a snap-on 30 mm cage bracket (CPMA3, Thorlabs) with an M4 cap screw, then clamp the bracket to a fully seated Ø12 mm post (TR50/M, Thorlabs) so that the bracket and holder remain in direct contact.

31. Attach a 30 mm cage-cube polarizing beamsplitter (CCM1-PBS251/M, Thorlabs) to the bracket using four 50.8 mm Ø6 mm cage rods (ER2, Thorlabs) along the through axis and a second set of four ER2 rods on the orthogonal face of the cube.

32. Fasten an SM1-to-fiber adapter module (SM1FCA, Thorlabs) mounted on a 30 mm SM1 cage plate (CP33/M, Thorlabs) to each port of the cube, leaving approximately 100 mm between each single-mode fiber tip and the lens center to allow dual-input excitation **(Fig. 1g)**.

33. Fold the beam with an elliptical dielectric mirror (BBE1-E02, Thorlabs) held in a right-angle kinematic cage mount (KCB1EC/M, Thorlabs), linking this mount to a second CP33/M plate via Ø6 mm cage rods totaling 228.6 mm (ER4 + ER3 + ER2) to yield a plate-to-mirror distance of approximately 195 mm that accommodates the fixed $f = 500$ mm cylindrical-lens position **(Fig. 1h,i: h, intermediate assembly; i, components arranged on a breadboard)**.

34. Translate the entire assembly until the focused beam coincides with a mark located 500 mm upstream of the sample center on the objective axis, then tighten all M4 and M6 fasteners **(Fig. 1j: Completed setup)**.

Most structural components in descSPIM-Galaxy, except for the detection optics and several excitation subassemblies, match those used in descSPIM-basic. Therefore, consult the detailed construction workflow described in the descSPIM-basic procedures (ver. 230520;

https://github.com/dbsb-juntendo/descSPIM/blob/main/descSPIM-basic/descSPIM_construct

[ion_ver230520.pdf](#)) in parallel during this step.

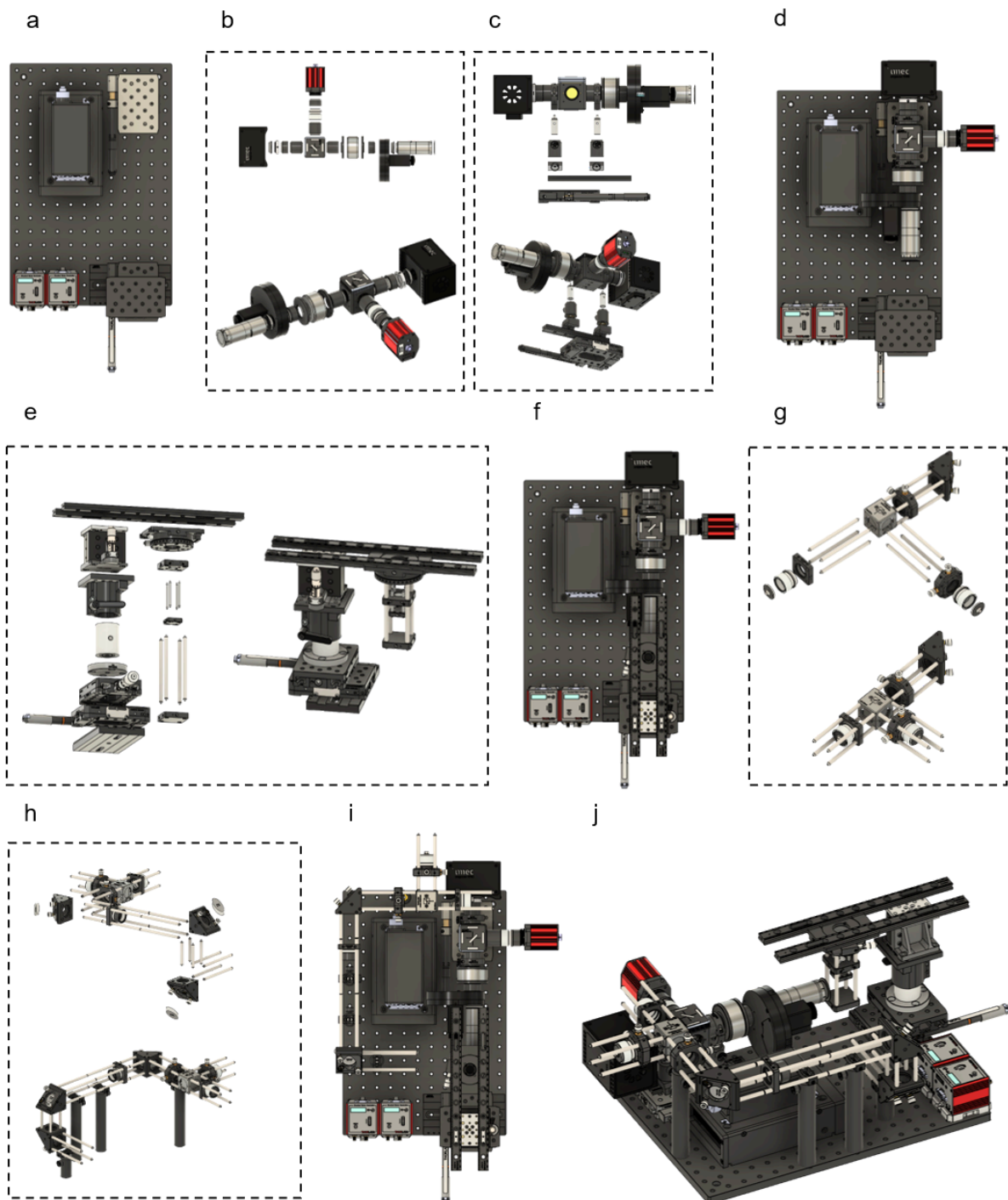


Fig. 1 | Graphical illustration of the building-up process of descSPIM-Galaxy

PAUSE POINT: After completing microscope construction, pause the protocol here before optical alignment. Secure all components, switch off all laser sources, and cover the setup to prevent dust accumulation. The assembled setup can be maintained under standard laboratory conditions, and alignment can be resumed at any time.

Optical alignment of descSPIM-Galaxy (see supplementary fig.6)

Timing 2 h-4 h

35. Before integrating the two laser combiners, perform the optical alignment according to the *Optical alignment procedures of descSPIM (ver. 230520)* (available at https://github.com/dbsb-juntendo/descSPIM/blob/main/descSPIM-basic/descSPIM_opticalalignment_ver230520.pdf). Since the descSPIM-Galaxy shares the same fundamental optical architecture as the descSPIM-basic, you can apply the alignment procedure described in that document directly to the Galaxy system as well.
36. Prepare a 1% (w/v) agarose solution in PBS by mixing agarose (Nacalai Tesque, cat. no. 01163-76) with phosphate-buffered saline (PBS; Nacalai Tesque, cat. no. 14249-24).
37. Loosely cap the tube and heat the mixture in a microwave oven until it boils, then stop heating.
38. Repeat the heating step at least twice to ensure complete dissolution of the agarose.
39. Place the melted agarose–PBS solution in a 70 °C water bath (square shape, liquid expansion type for stirrer; Advantec, cat. no. TBS181SB) for ≥5 min to remove bubbles.
40. In a microtube, combine 15 µL Protein A/G PLUS–Agarose (Santa Cruz Biotechnology, cat. no. sc-2003) with 2 µg Goat anti-Mouse IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 405 (Abcam, cat. no. ab175659), and 2 µg Goat anti-Mouse IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 488 (Abcam, cat. no. ab150113).
41. Bring the total volume to 40 µL with PBS, mix gently, and incubate the mixture at 4 °C for 1 h.
42. Wash the agarose beads three times for 1 min each with cold PBS at 4 °C, then adjust the final PBS volume to 100 µL.
43. Gently mix the bead suspension (100 µL) with 900 µL of 1% agarose in PBS prepared above by careful pipetting to avoid introducing bubbles, then pour the

mixture into a four-sided transparent cuvette (borosilicate glass, 10 mm or 20 mm path length; GL Science, cat. no. F10-G-10 or F10-G-20) or a plastic cuvette (Bio-Rad, cat. no. 1702415).

44. Place the cuvette on a 4 °C cooling block and leave it undisturbed for approximately 10 min to allow the agarose to set and residual bubbles to dissipate.
45. Insert the silver mirror and mount the kinematic cage cube (DFM1/M + PFR10-P01, Thorlabs), then excite the sample using 405 nm and 488 nm excitation.
46. Using the CMOS camera (CS895MU, Thorlabs), focus at an arbitrary Z position and acquire two image stacks (405 nm and 488 nm excitation) with 20 µm axial steps (approximately 30 slices each).
47. In ImageJ, match the pixel sizes of the two stacks and use *Image* → *Color* → *Merge Channels* to generate a dual-color volume.
48. Inspect orthogonal (XZ and YZ) views to assess any Z-offset between channels.
49. If misalignment persists, adjust the sample position using the XY translator (CXY1A, Thorlabs) and repeat image acquisition until the channels are colocalized.

Caution: Do not overheat, as the solution may boil over.

Critical Step: Incomplete dissolution can leave residues that obstruct excitation light during imaging.

Critical Step: Ensure no visible bubbles remain before use.

2. Sample Preparation

Cleared tissue preparation

7-Color Immunostaining and Clearing of E12.5 Mouse Embryo for descSPIM-Galaxy Imaging

Sample Preparation

Timing 2 days

50. Euthanize timed-pregnant ICR mice by CO₂ inhalation in accordance with institutional animal care and use guidelines.
51. Open the abdominal cavity via laparotomy and harvest E12.5 embryos under a stereomicroscope; do not perfuse.
52. Immerse-fix isolated embryos in freshly prepared 4% paraformaldehyde (PFA) in PBS (prepared from paraformaldehyde granular; Nacalai Tesque, cat. no. 02890-45; ≥90% as formaldehyde) at 4 °C for 24 h.
53. Wash embryos three times in PBS, 1 h each at 4 °C.

Delipidation and oxidative decolorization

Timing 3 days

54. Pre-equilibrate in 50% (v/v) CUBIC-L (Tokyo Chemical Industry Co., Ltd. (TCI), cat. no. T3740) for 1 h at room temperature.
55. Incubate in 100% CUBIC-L for 24 h at room temperature to achieve delipidation.
56. Wash in PBS for 2 h at room temperature to remove residual reagent.
57. Incubate in 3% hydrogen peroxide (H₂O₂), 35% w/v solution (Nacalai Tesque, cat. no. 20779-65), diluted 1:10 with Milli-Q water (Millipore, ultrapure grade) and containing 150 mM sodium chloride (NaCl) (Nacalai Tesque, cat. no. 31320-05) at 37 °C for 48 h to inactivate endogenous peroxidases and oxidize blood-derived heme/iron.
58. Wash in PBS for 2 h at room temperature.

Immunostaining

Timing 3 days

59. Pre-incubate embryos overnight in 10 mL of 5% methanol (Nacalai Tesque, cat. no. 21915-93) in PBS at 37 °C. Pre-incubate embryos in H-TSC buffer (to be provided later) at 37 °C for 1 h.
60. Transfer samples into 2 mL round-bottom cryo vials (Greiner Bio-One, cat. no. 121263), immerse in staining buffer, and seal under a layer of mineral oil (to be provided later) at room temperature.
61. Prepare a staining solution with a total volume of 100 μ L by combining 50 μ L of 2 $\times$ H-TSC supplemented with 3% (v/v) HV-Additive (#0854, CUBIC Histovision 2.0; CUBIC stars) to yield a final HV-Additive concentration of 1.5% after dilution to 1 $\times$.
62. Add 2 μ L each of six primary antibodies (approximately 1 μ g/ μ L) that have been pre-labeled according to the manufacturers' instructions as follows: TAG-1 (R&D Systems, cat. no. AF4439) labeled using a FITC Conjugation Kit (Lightning-Link®, Abcam, cat. no. ab102884); TNNI3 (Abcam, cat. no. ab56357) labeled using a Cy3® Conjugation Kit (Fast) (Lightning-Link®, Abcam, cat. no. ab188287); TUBB3 (Covance, cat. no. MMS-435P) labeled using an Alexa Fluor® 594 Conjugation Kit (Fast) (Lightning-Link®, Abcam, cat. no. ab269822); RUNX2 (Abcam, cat. no. ab192256) labeled using an Alexa Fluor® 647 Conjugation Kit (Fast) (Lightning-Link®, Abcam, cat. no. ab269823); α -SMA (Sigma-Aldrich, cat. no. A5228) labeled using a Cy5.5® Conjugation Kit (Fast) (Lightning-Link®, Abcam, cat. no. ab269891); and CD31 (R&D Systems, cat. no. AF3628) labeled using a DyLight® 755 Conjugation Kit (Lightning-Link®, Abcam, cat. no. ab269890).
63. Add 6 μ L Oxazole Blue Homodimer (POPO-1) (Biotium, cat. no. 40093) and 32 μ L Milli-Q water (Millipore, ultrapure grade) to bring the total volume to 100 μ L, and mix gently.
64. Incubate the sample in the staining solution at room temperature for 3 d, then continue incubation at 4 °C for an additional 1 d.

Critical Step: Follow the antibody concentration recommended in the Lightning-Link® labeling kit protocol. In this protocol, all labeled antibodies are used at a uniform final

concentration of 1.0 mg mL⁻¹; therefore, if the original stock concentration differs from 1.0 mg mL⁻¹, concentrate the antibody to ~1.0 mg mL⁻¹ at 4 °C using Amicon® Ultra 30 kDa MWCO centrifugal filters prior to labeling.

65. Wash in H-TS three times, 1 h each at 4 °C.

66. Post-fix embryos in 1% formaldehyde solution (prepared from Formaldehyde solution, 36–38% (w/v); Nacalai Tesque, cat. no. 16223-55) in H-TS overnight at 4 °C, then fix again in 1% formaldehyde solution at 37 °C for 1 h. Wash in PBS for 2 h at room temperature.

RI Matching

Timing 3 days

67. Pre-equilibrate in 50% (v/v) CUBIC-R (Tokyo Chemical Industry Co., Ltd. (TCI), cat. no. T3983) for 24 h at room temperature.

68. Incubate in 100% CUBIC-R for 48 h at room temperature to reach the final refractive index.

Embedding

Timing 1 h

69. Prepare a four-sided transparent cuvette—either a borosilicate glass type, 10 mm or 20 mm path length (GL Science, cat. no. F10-G-10 or F10-G-20), or a plastic type, 10 mm path length (Bio-Rad, cat. no. 1702415)—and prepare 2% (wt/vol) agarose dissolved in 100% CUBIC-R.

70. Pour approximately 1 ml of the agarose solution into the cuvette. Lay the cuvette on its side and seal the opening with Parafilm (or an equivalent film) to prevent leakage.

71. Place the cuvette on a cooling block and incubate it for 4–5 min until the agarose has completely solidified.

72. Place the cleared embryo gently on top of the solidified agarose surface.

73. Pour the remaining agarose solution into the cuvette to completely cover the sample.

74. Keep the cuvette upright and incubate it on the cooling block for approximately 10 min to allow complete gel solidification.

PAUSE POINT: After embedding, store the sealed sample at room temperature in the dark. Under these conditions, the embedded sample remains stable for up to 2–3 months, and imaging can be resumed without additional preparation.

3. Image acquisition

Timing 3 days

75. For each dye (RUNX2–Alexa Fluor 647, α -SMA–Cy5.5, TNNI3–Cy3, and TUBB3–Alexa Fluor 594), mount the corresponding single-stained sample on the microscope stage.
76. Set the laser power and exposure time to match the settings planned for multiplex image acquisition.
77. Launch the HSI Snapscan application and select the excitation wavelength corresponding to the dye being measured.
78. Identify the field of view that exhibits the highest fluorescence signal.
79. Crop a region of interest that contains only the fluorescence signal.
80. Export the automatically computed mean fluorescence spectrum and save it as the reference spectral profile.

CRITICAL STEP: It is possible to apply the optimized laser power and exposure settings—which you determine on single-stained controls—once to the seven-colour multiplex sample using the HSI Snapscan application to verify relative signal intensities. However, because each test acquisition exposes the same region to approximately 3 minutes of continuous laser illumination, extensive or repeated HSI Snapscan imaging can induce photobleaching; therefore, avoid extensive or repeated HSI Snapscan imaging.

CRITICAL STEP: Perform image acquisition using a custom control environment integrating Thorlabs Kinesis® and ThorCam® for the descSPIM-Galaxy system. Synchronize the sample-holder stage and the detection-camera stage via Kinesis. Set the synchronous speed correction factor between the two actuators to 0.342⁴⁰, yielding a velocity ratio of 1 : 0.342 (e.g., 1 mm s⁻¹ : 0.342 mm s⁻¹) for sample vs. camera stages.

81. Insert a 45° silver mirror into the kinematic cage cube (KCB1EC/M, Thorlabs) to enable CMOS detection. Mount the embedded cleared sample onto the sample holder. Turn on the laser combiner and launch ThorCam® in live-view mode. Using the live image and Kinesis® stage controls, adjust both stages so the focal plane is centered in the specimen. Set exposure to 300 ms for all channels.

82. Open Spyder and run `Laser and Filter Wheel Controller_GUI.py` in `dbsb_juntendo/descSPIM/descSPIM-Galaxy/` (GitHub). The GUI synchronizes the laser-combiner output and the motorized filter wheel. An equivalent `.ipynb` exists, but for long-term stability we recommend the local `.py`. Launch Kinesis® (Z825B actuators) and ThorCam® (CS895MU camera).

83. In ThorCam, open *Settings* → *Hardware Triggering* and set Mode = Standard, Frames per Trigger = 1, and Polarity = On High.

84. In ThorCam, click *Acquire*, choose the output folder and file name, verify laser power and filter-wheel position, then begin the z-stack acquisition. Acquire three channels sequentially under the following conditions: CD31–DyLight 755: 730 nm, 50 mW, 750 nm long-pass, 300 ms; TAG-1–Alexa 488: 488 nm, 20 mW, 520/40 nm band-pass, 300 ms; POPO-1: 405 nm, 20 mW, 425 nm long-pass, 300 ms.

CRITICAL STEP: descSPIM-Galaxy uses stepwise z-stack (step-and-acquire). The stage moves in 20 µm steps and acquires one image per step. Before imaging, open the `.sequence` file (e.g., `Galaxy_imaging.sequence`) and confirm: (i) *Edit* → *Repeat* → Repeat Count = 300 for each channel; (ii) *Parameters* → Move Relative (at velocity): set to Enabled; Move Distance = 20 µm. (iii) total travel = 20 µm × 300 = 6 mm, ensuring uniform 20 µm spacing and ~300 sections per channel.

85. After verification, press *REC* in ThorCam to set the file name, then start acquisition by executing *RUN* in Kinesis. Set image bit depth to 8-bit to minimize I/O delays. Upon completion of each stack, stages return to the stored z-position. After completing each stack, control subsequent channel switching (laser line and emission filter) within `Laser and Filter Wheel Controller_GUI.py`.

CRITICAL STEP: Do not adjust or touch any optical components outside of the defined acquisition controls. In particular, avoid disturbing the sample holder, as any movement will invalidate the reproducibility of both CMOS and hyperspectral acquisitions. Likewise, do not tamper with the polarizing beamsplitter assembly, since misalignment here will disrupt the beam paths from the Cobolt Skyra and C-FLEX sources.

86. After completing CMOS-mode imaging, record the current sample and camera stage Z-coordinates as `sample_start_Z` and `camera_start_Z` (“start position”).

87. Remove the 45° silver mirror from the kinematic cage cube. Close ThorCam™, Kinesis®, and exit HSI Snapscan.
88. Using only the sample stage, translate the sample to a position approximately at the specimen's center where the laser still illuminates the tissue; record this Z as *sample_focus_Z*.
89. Launch HSI Snapscan and select the first excitation line.
90. Keep the laser and filter settings unchanged and use the live spectral view to bring the fluorescence into sharp focus.
91. Using Kinesis®, move only the camera stage to maximize focus and record the corresponding Z position as *camera_focus_Z*.
92. Repeat steps 1–3 for each excitation wavelength (for example, 647 nm and 561 nm), and move only the camera stage for all adjustments.
93. Compute each excitation's true camera-start Z by:
$$\text{camera_start_Z} = \text{camera_focus_Z} - 0.342 \times (\text{sample_focus_Z} - \text{sample_start_Z}).$$
94. In Spyder, open and run [Snapscan_Multiex_3D_GUI.py](#) (dbsb_juntendo/descSPIM/descSPIM-Galaxy/) to launch the multicolour hyperspectral 3D-imaging GUI. Enter each calculated *camera_start_Z* into its corresponding excitation channel field.
95. With ThorCam®, Kinesis® and HSI Snapscan closed, click *Start* in the [Snapscan_Multiex_3D_GUI.py](#) GUI. Click Start in the Snapscan_Multiex_3D_GUI.py GUI to run the hyperspectral acquisition automatically through your predefined excitation lines, filters, exposure times, and 20 µm z-steps to generate the full multiplexed 3D dataset. **Note: From the GUI, freely specify both the exposure time and the sample step (300 ms and 20 µm in this example).**

5. Image Processing and Registration

Post-acquisition data preparation

Timing 30 min - 1 h

96. Open Jupyter Notebook and launch `Endmember_and_BandExtraction_Combined.ipynb`.
97. In the first cell (`preprocess_endmember_gui_single_xml`), select the XML file once.
98. Add the measured CSV spectra for TNNI3–Cy3 and TUBB3–Alexa 594 (ex561), and for RUNX2–Alexa 647 and aSMA–Cy5.5 (ex647), and use “aSMA” instead of “αSMA” to avoid path recognition errors.
99. Run the cell to generate the following files:
`Interp_crop_TNNI3-Cy3_loopX_ex561nm.csv`,
`Interp_crop_TUBB3-Alexa594_loopX_ex561nm.csv`,
`Interp_crop_RUNX2-Alexa647_loopX_ex647nm.csv`, and
`Interp_crop_aSMA-Cy5.5_loopX_ex647nm.csv`.
100. In the next cell (`extract_valid_bands_gui.py`), select an example loop TIFF file for each channel to process (for example, the `loop1` file in each excitation folder).
101. Set the Output root folder and run the cell.
102. Allow the tool to automatically detect and process all matching `loop{wl}nm_objectms.tif(f)` files in the same folder and save the results as `Extracted_<original_name>` under `Extracted_ex{wl}`.
103. If necessary, adjust the long-pass wavelength ranges in the GUI (for example, 575–750 nm for ex561 and 675–850 nm for ex647) before running the extraction.
104. After the script finishes, verify that the script successfully processed all input endmember CSVs and all input mixed data. Once both ex561 and ex647 preprocessing are complete, proceed to linear unmixing.

Image processing

Timing 2 day - 4 days

105. Launch [Unmix_Gui_Multi ex.ipynb](#), set the input directories for your extracted TIFFs (e.g., [Extracted_loopX_561nm_object_300ms.tif](#), [Extracted_loopX_647nm_object_300ms.tif](#), ...), add the corresponding endmember CSVs in the GUI, and choose the unmixing algorithm (NNLS or FCLSU). Run to automatically generate unmixed TIFF outputs for each excitation channel in a single session. For example:
[Cy3/A594_Extracted_loop12_561nm_object_300ms.tif](#),
[Cy5.5/Runx2_Extracted_loop12_647nm_object_300ms.tif](#)
106. Open the Jupyter notebook [Stack_and_Avg_GUI.ipynb](#) and set the Output root folder.
107. For ex561 data, set the input directory to the folder containing files such as [Cy3_Extracted_loopX_561nm_object_300ms.tif](#) or [A594_Extracted_loopX_561nm_object_300ms.tif](#).
108. Allow the program to automatically detect the label name from the prefix before the first underscore (for example, Cy3 or A594) and generate the corresponding stack files (for example, [Cy3_loop1–N_stack.tif](#) and [A594_loop1–N_stack.tif](#)), with optional per-loop single images saved in subfolders.
109. For ex647 data, set the input directory to the folder containing files such as [Label_Extracted_loopX_647nm_object_300ms.tif](#).
110. Allow the program to generate per-label stack files (for example, [Label_loop1–N_stack.tif](#)).
111. For ex405 data, set the input directory to the raw cube folder and specify channels 1–5 for averaging.
112. Allow the program to generate [ex405_avg_loopX.tif](#) files and assemble them into a single [ex405_avg_stack.tif](#) file.

113. Click **Run** once to process all configured inputs and generate all output files in a single batch.
114. Before starting, save the input CMOS TIFF stacks directly from Imaging Protocol 61 using filenames that include the dye identity (for example, POPO1_CMOS.tif, TAG1_CMOS.tif, and CD31_CMOS.tif).
115. Preserve these filenames throughout processing so that the workflow automatically propagates the dye identity into the downsampled and registered output files.
116. Open the Jupyter notebook `tiff_Downsample_and_ECC_Registration_GUI`.
117. Run the first notebook cell to launch the downsampling GUI for CMOS TIFF stacks (for example, POPO-1, TAG-1-FITC, and CD31-D755).
118. Select the input TIFF stacks and an output folder, and set the downsampling parameters to convert the pixel size from 3.45 $\mu\text{m}/\text{px}$ to 5.5 $\mu\text{m}/\text{px}$.
119. Generate downsampled files, which the program saves with filenames such as POPO1_CMOS_binned.tif, TAG1_CMOS_binned.tif, and CD31_CMOS_binned.tif.
120. Run the second notebook cell to launch the ECC registration GUI.
121. Set `ex405_avg_stack.tif` as the fixed reference image and load all downsampled CMOS stacks as the moving images.
122. Perform rigid affine registration and generate aligned output files such as POPO1_CMOS_rigid_affine.tif, TAG1_CMOS_rigid_affine.tif, and CD31_CMOS_rigid_affine.tif.
123. Run each notebook cell separately and follow the GUI prompts to complete file selection and processing.

124. Upload `ex405_avg_stack.tif` and the rigid/affine-registered stacks to your Linux server, run **ANTS Registration GUI.py**, and generate and verify `POP01_CMOS_SyN.tif`, `TAG1_CMOS_SyN.tif`, and `CD31_CMOS_SyN.tif`.

125. In ImageJ, open all seven stacks. For each stack, input the xy pixel size and z depth from its properties. Convert all stacks to 8-bit TIFF. Next, choose Image → Color → Merge Channels, assign a unique LUT (Look-Up Table) to each channel, and save the result as `7color_merged.tif`. Finally, run Image → Stacks → Z Project Max Intensity and save the output as `7color_MIP.tif`. This procedure enables the generation of visually distinct multi-channel images and maximum intensity projections.